Chapter 10 - Markov Chain Monte Carlo: Exercise solutions

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Click on the arrow to see a solution.

Exercise 10.1

This exercise extends the analysis of the Weibull model for survival times of lung cancer patients in Exercise 10.1 to the regression setting. The survival times are here modelled as independent Weibull distributed with a scale parameter λ that is a function of covariates, i.e. using a Weibull regression model. The response variable `time` is denoted by y and is modelled as a function of the three covariates `age`, `sex` and `ph.ecog` (ECOG performance score). The model for patient i is:

$$y_i|\mathbf{x}_i,\beta,k \overset{\text{ind}}{\sim} \text{Weibull}(\lambda_i = \exp(\mathbf{x}_i^\top\beta), k).$$

where β is the vector with regression coefficients. Note that by the properties of the Weibull distribution, the conditional mean in this model is $\mathbb{E}(y|\mathbf{x}_i) = \lambda_i\Gamma(1+1/k)$, so the regression coefficients do not quite have the usual interpretation of the effect on the conditional mean. The three covariates are placed in a $n \times p$ matrix $\mathbf{X}$ with the first column being one for all observations to model the intercept. Use a multivariate normal prior for $\beta \sim N(\mathbf{0}, \tau^2\mathbf{I}_p)$ with the non-informative choice $\tau = 100$, and the prior $k \sim \text{logNormal}(0, 2^2)$. Remove the patients with missing values in the selected covariates.

Sample from the posterior distribution $p(\beta, k|\mathbf{y}, \mathbf{X})$ using HMC in stan. Plot the marginal posterior for k and the marginal posteriors of each regression coefficient.

Solution

```
library(tidyverse) # loads data manipulation and visualization packages
library(rstan)
colors = c("#6C8EBF", "#c0a34d", "#780000","#007878","#B5C6DF","#EADAAA","#AE6666")
```

The code below defines the iid Weibull survival model with censored data in `stan`. The code here extends this example in the Stan User Guide to the case with *different censoring*

points for each patient. Note the `target +=` construction where the censored data points are added to the target (the log posterior) after the initial uncensored (observed) data are included in the log posterior with the `y_obs ~ weibull(k, lambda)` statement. The `weibull_lccdf` function in stan is a convenience function that computes the survival probability $\Pr(X >= x) = 1 - F(x)$, where $F()$ is the cdf of the Weibull distribution. There are `_lccdf` versions of all distribution in stan.

```
weibull_survivalmodel <- '
data {

  // Data
  int<lower=0> N_obs;
  int<lower=0> N_cens;
  array[N_obs] real y_obs;
  array[N_cens] real y_cens;

  // Model setting
  real<lower=0> k;

  // Prior hyperparameters theta ~ Gamma(alpha, beta)
  real<lower=0> alpha;
  real<lower=0> beta;
}
parameters {
  real lambda;
}
model {
  lambda ~ gamma(alpha, beta); // specifies the prior
  y_obs ~ weibull(k, lambda);  // add the observed (non-censored) data
  target += weibull_lccdf(y_cens | k, lambda); // add censored. lccdf is 1-cdf
}
'
```

We set up the data and prior lists that will be supplied to stan:

```r
library(survival) # loads the lung cancer data as `lung`
k = 3/2
y_obs <- lung %>% filter(status == 2) %>% pull(time)
y_cens <- lung %>% filter(status == 1) %>% pull(time)

data <- list(N_obs = length(y_obs), N_cens = length(y_cens),
             y_obs = y_obs, y_cens = y_cens, k = k)

alpha_prior <- 3     # shape parameter
beta_prior <- 1/50   # rate parameter
prior <- list(alpha = alpha_prior, beta = beta_prior)
```

Load rstan and set some options

```r
#install.packages("rstan", repos = c('https://stan-dev.r-universe.dev',
#                                    getOption("repos")))
suppressMessages(library(rstan))
options(mc.cores = parallel::detectCores())
rstan_options(auto_write = TRUE)
```

Sample from the posterior distribution using HMC in stan

```r
nDraws = 5000
fit = stan(model_code = weibull_survivalmodel, data = c(data, prior), iter = nDraws)
```

Exercise 10.2

Exercise 2.10 plotted the posterior for the parameter λ in the Weibull model

$$X_1, \ldots, X_n | \lambda, k \overset{\text{iid}}{\sim} \text{Weibull}(\lambda, k).$$

for different fixed values of k. This exercise asks you to implement the same model in `stan` to sample from the posterior distribution of λ for a given $k = 3/2$. The Stan User Guide describes how to implement censoring in the model. The example in the User Guide has the same censoring point for all patients, which is not the case in the `lung` dataset. So you need to generalize that to a vector of censoring points, one for each patient.

Solution

```
library(tidyverse) # loads data manipulation and visualization packages
library(rstan)
colors = c("#6C8EBF", "#c0a34d", "#780000","#007878","#B5C6DF","#EADAAA","#AE6666")
```

Set data and set up prior hyperparameters

```
library(survival) # loads the lung cancer data as `lung`
lung <- lung %>% select(c("time", "status", "age", "sex", "ph.ecog")) %>% drop_na()
y = lung$time
X = cbind(1, lung$age, lung$sex == 2, lung$ph.ecog) # sex = 1 is female
p = dim(X)[2]
censored = (lung$status == 1)
y_obs = y[-censored]
y_cens = y[censored]
X_obs = X[-censored,]
X_cens = X[censored,]
mu <- rep(0,p)  # beta ~ N(mu, tau^2*I)
tau <- 100
mu_k <- 0       # k ~ LogNormal(mu_k, sigma_k^2)
sigma_k <- 2
```

Set up the stan model for the Weibull regression.

```
weibull_survivalreg <- '
data {

  // Data
  int<lower=0> N_obs;
  int<lower=0> N_cens;
  int<lower=1> p;
  array[N_obs] real y_obs;
  array[N_cens] real y_cens;
  matrix[N_obs,p] X_obs;
  matrix[N_cens,p] X_cens;

  // Prior hyperparameters k ~ LogNormal(mu_k, sigma_k) and beta_ ~ N(0, tau^2*I)
  real mu_k;
  real<lower=0> sigma_k;
  real<lower = 0> tau;
}
parameters {
  vector[p] beta_;
  real<lower=0> k;
}
model {
  k ~ lognormal(mu_k, sigma_k);    // specifies the prior
  beta_ ~ normal(0, tau);
  y_obs ~ weibull(k, exp(X_obs * beta_));  // add the observed (non-censored) data
  target += weibull_lccdf(y_cens | k, exp(X_cens * beta_)); // add censored.
}
'
```

We set up the data and prior lists that will be supplied to stan:

```
data <- list(p = dim(X_obs)[2], N_obs = length(y_obs), N_cens = length(y_cens),
             y_obs = y_obs, y_cens = y_cens, X_obs = X_obs, X_cens = X_cens)

prior <- list(tau = tau, mu_k = mu_k, sigma_k = sigma_k)
```

Load rstan and set some options

```
#install.packages("rstan", repos = c('https://stan-dev.r-universe.dev',
#                                     getOption("repos")))
suppressMessages(library(rstan))
options(mc.cores = parallel::detectCores())
rstan_options(auto_write = TRUE)
```

Sample from the posterior distribution using HMC in stan

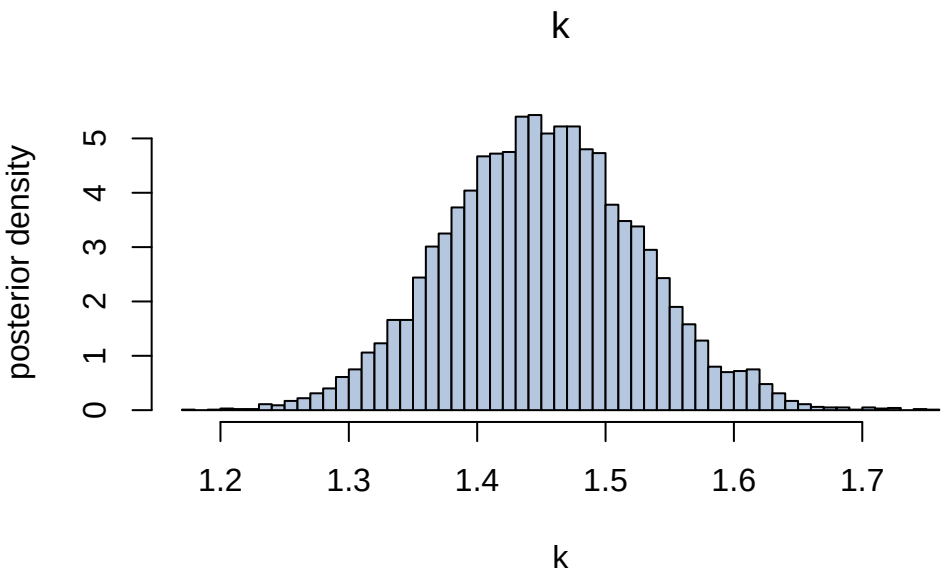
```
nDraws = 5000
fit = stan(model_code = weibull_survivalreg, data = c(data, prior), iter = nDraws)
```

```
s <- summary(fit, pars = c("beta_", "k"), probs = c(0.025, 0.975))
s$summary  # results from all the different runs (chains) merged.
```

```
                mean      se_mean          sd        2.5%        97.5%
beta_[1]  6.323585021 4.804896e-03 0.321527311  5.69876268  6.954556174
beta_[2] -0.002785287 7.750432e-05 0.005193242 -0.01295855  0.007288999
beta_[3]  0.235543271 1.122063e-03 0.095764287  0.05092624  0.431659872
beta_[4] -0.260410649 7.916868e-04 0.069295943 -0.39506230 -0.128464942
k         1.451221040 8.791999e-04 0.074864859  1.30662177  1.604521677
           n_eff      Rhat
beta_[1] 4477.831 1.0000759
beta_[2] 4489.783 1.0000969
beta_[3] 7284.045 0.9998708
beta_[4] 7661.413 1.0001000
k        7250.714 0.9999998
```

Plotting the marginal posterior of k

```
# Plot histogram from stan draws
postsamples <- extract(fit, pars = c("beta_","k"))
hist(postsamples$k, 50, freq = FALSE, col = colors[5],
     xlab = expression(k), ylab = "posterior density",
     main = expression(k))
```



Plotting the marginal posteriors for each β coefficient

```
varNames = c("intercept", "age", "sex", "ph.ecog")
par(mfrow = c(2,2))
for (j in 1:p){
  hist(postsamples$beta_[,j], 50, col = colors[6],
       xlab = expression(beta), ylab = "density", main = varNames[j])
}
```

